

Multi-Robot Tracking in Indoor Service Environments

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Outline

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- Autonomous service robots are increasingly used in indoor environments.
- Accurate localization is essential for safe operation.
- Sensor measurements are often affected by noise and temporary failures.
- Reliable tracking methods are required.

Project Objectives

- Develop a multi-robot tracking framework.
- Simulate a restaurant environment in Webots.
- Apply Kalman Filter based tracking.
- Investigate self-healing tracking under missing measurements.
- Compare Kalman Filter and Extended Kalman Filter (EKF).

Tracking Challenges

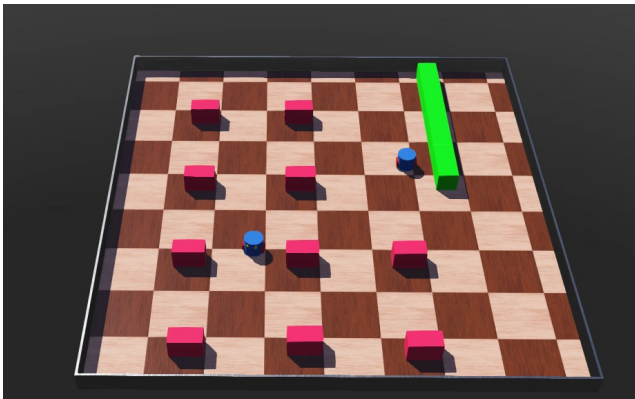
- Measurement Noise
- Missing Measurements
- Multiple Moving Robots
- Real-Time Constraints

Objective

Develop a robust tracking system capable of maintaining accurate position estimates despite sensor noise and temporary measurement outages.

Simulation Environment

- Restaurant environment created in Webots.
- Arena size: 2 m $\times$ 2 m.
- Two autonomous service robots.
- Ten customer tables and one food stand.
- Noisy position measurements generated by a supervisor node.



State-Space Model

$$\mathbf{x}_k = \begin{bmatrix} p_x \\ p_z \\ v_x \\ v_z \end{bmatrix}$$

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + w_k$$

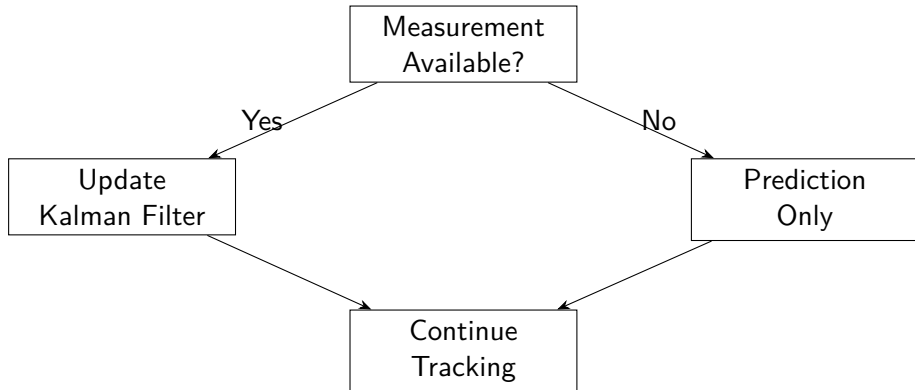
- Constant velocity motion model.
- Position and velocity states.
- Gaussian process noise.

- Prediction step
- Covariance propagation
- Kalman gain computation
- Measurement update

$$\hat{x}_{k|k-1} = F\hat{x}_{k-1|k-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$

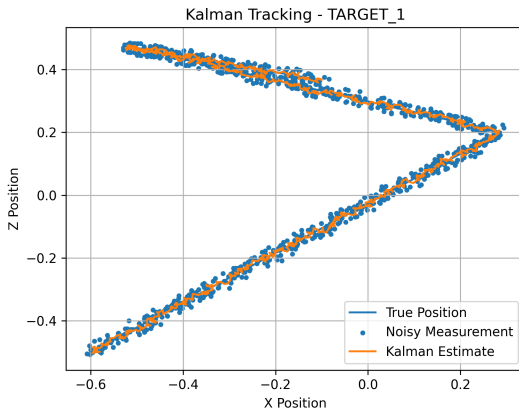
Self-Healing Tracking Mechanism



- Measurements available → standard Kalman update.
- Measurements unavailable → prediction step only.
- Continuous tracking maintained during outages.

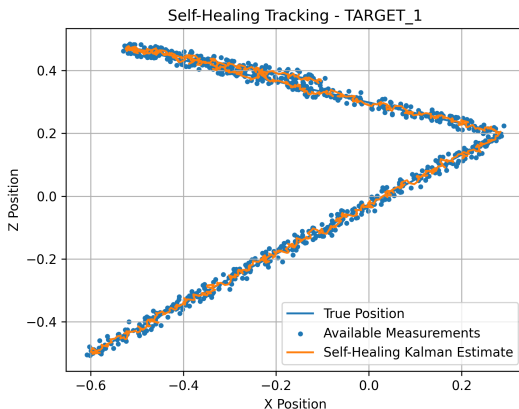
Kalman Filter Results

Target	Measurement RMSE	KF RMSE
TARGET_1	0.0117	0.0060
TARGET_2	0.0115	0.0059



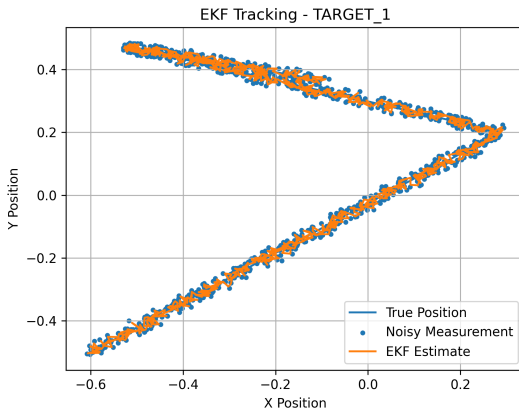
Self-Healing Results

Target	Loss Rate	Measurement RMSE	SH RMSE
TARGET_1	16.67%	0.0117	0.0065
TARGET_2	15.51%	0.0114	0.0062



EKF Comparison

Target	Measurement	KF	EKF
TARGET_1	0.0117	0.0060	0.0095
TARGET_2	0.0115	0.0059	0.0092



Conclusions

- Kalman Filter reduced tracking error by approximately 48
- Self-healing mechanism remained effective despite measurement losses.
- EKF improved measurements but did not outperform KF.
- Linear KF was sufficient for the simulated robot trajectories.

Thank You